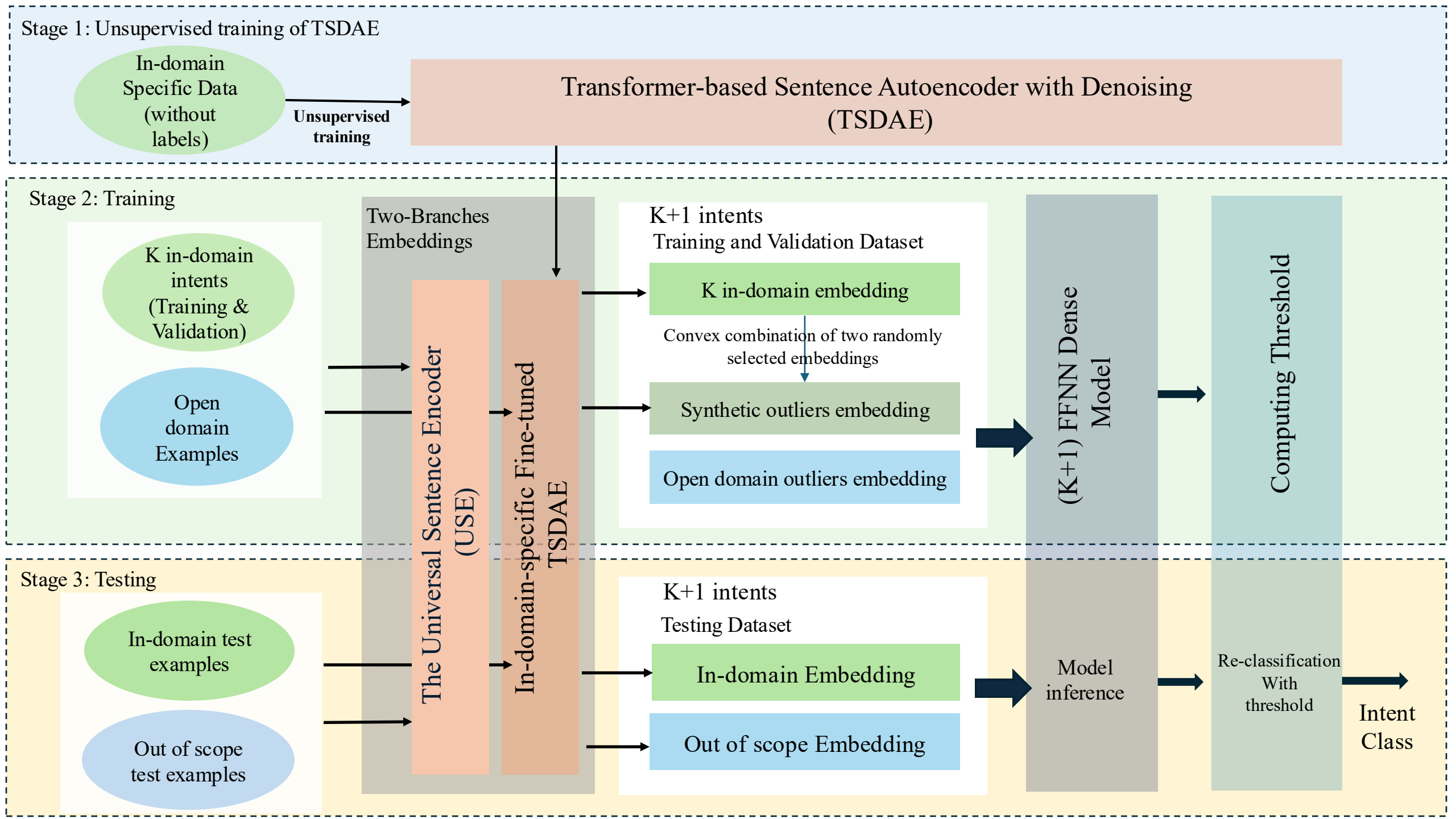
Stage 1: Unsupervised training of TSDAE
In-domain Specific Data (without labels)
Unsupervised training
Transformer-based Sentence Autoencoder with Denoising (TSDAE)
Stage 2: Training
K in-domain intents (Training & Validation)
Open domain Examples
Two-Branches Embeddings
The Universal Sentence Encoder (USE)
In-domain-specific Fine-tuned TSDAE
K+1 intents
Training and Validation Dataset
K in-domain embedding
Convex combination of two randomly selected embeddings
Synthetic outliers embedding
Open domain outliers embedding
(K+1) FFNN Dense Model
Computing Threshold
Stage 3: Testing
In-domain test examples
Out of scope test examples
K+1 intents
Testing Dataset
In-domain Embedding
Out of scope Embedding
Model inference
Re-classification With threshold
Intent Class